# Video Object Tracking on MOT16

Faster R-CNN detection with Siamese re-identification

Ben Blake  Geethika Padamati  Tina Nguyen  Sanjana Aileni

Midterm Presentation

Group 4
COMP-SCI 5567, Deep Learning
University of Missouri–Kansas City

# Team

**Ben Blake**, detection and tracking
Faster R-CNN, data association, integration

**Geethika Padamati**, data pipeline
MOT16 parsing, datasets, augmentation

**Tina Nguyen**, re-identification
Siamese network, contrastive loss

**Sanjana Aileni**, evaluation and demo
Metrics, visualization, result video

# Problem

**Given** a video, **produce** a box around each pedestrian in every frame, with the same identity number following the same person.

### Detection is not enough

A detector answers "where are the people" separately in each frame. It has no way of knowing that the person in frame 10 is the person it saw in frame 9.

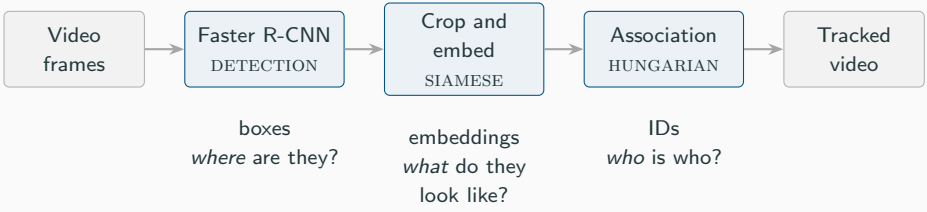
### Tracking adds identity

We have to link detections across frames, through occlusion, through people crossing, and through frames where the detector misses someone entirely.

### Why identity, and not just location

Driving, surveillance, sports and robotics all need to know *which* person, and for how long. Our target: track one pedestrian across an MOT16 sequence without losing them (rubric, grade A).

# System Architecture




**Offline tracking.** We process a batch of frames rather than streaming them, which lets us recover from short occlusions.

**Tracking by detection.** Detection and identity stay separate, so either component can be replaced without touching the other.

# Detection: Faster R-CNN

## Why this model

- The Region Proposal Network generates proposals inside the network, so no Selective Search
- Proposals and detection share one backbone, so each image is convolved once
- Good accuracy on small objects, and distant pedestrians are small

## Few-shot fine-tuning

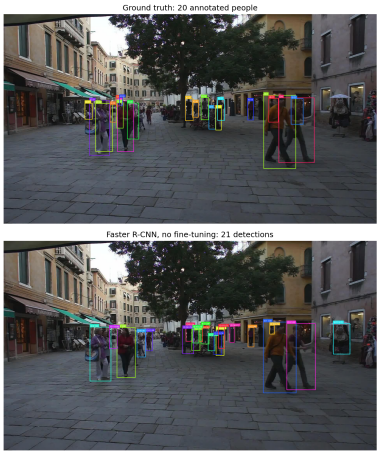
We freeze the ResNet50-FPN backbone and train only the detection heads, as the assignment suggests.

**Parameter budget**

| | |
|---|---|
| Total | 41.3 M |
| Frozen | 26.8 M |
| **Trainable** | **14.5 M** |

Training 35% of the parameters keeps the job inside Colab's memory budget.

# What the detector actually sees




Frame 1 of MOT16-02: ground truth above, our detections below. The detector finds the foreground and misses the small, distant and partly hidden people. That is where the missing 60% of recall goes.

# Association Problem

Each frame gives $N$ detections and we are holding $M$ tracks. Which detection continues which track?

### Overlap alone fails

When two people cross, each box overlaps the other's previous position, so position cannot separate them.

### Appearance alone fails

Two people in similar clothing produce similar embeddings, and appearance drifts as someone turns.

**Blended cost**

$$C_{ij} = (1-\lambda)\underbrace{(1 - \mathrm{IoU}_{ij})}_{\text{position}} + \lambda \underbrace{(1 - \cos(e_i, e_j))}_{\text{appearance}}$$

Minimised with the Hungarian algorithm, giving a globally optimal one-to-one assignment.

Pairs failing the IoU or appearance gate are rejected before matching. A track survives `max_age` unmatched frames, so a brief occlusion does not create a new identity.

# The MOT16 Dataset

- 14 sequences: 7 training with ground truth, 7 test
- Crowded streets, both moving and static cameras
- Heavy occlusion, with people leaving and re-entering the frame

### One annotation row

```
1, 3, 586, 447, 85, 263, 1, 1, 1
```

### Columns

| | |
|---|---|
| `frame` | 1 |
| `id` | 3 |
| `bb_left` | 586 |
| `bb_top` | 447 |
| `bb_width` | 85 |
| `bb_height` | 263 |
| `conf` | 1 (0 means ignore) |
| `class` | 1 (pedestrian) |
| `visibility` | 1.0 |

# Two traps in the annotations

**Coordinates are 1-indexed**

The files count pixels from 1 and NumPy counts from 0. If you miss this, every box is off by one pixel. The output still looks correct, but IoU drops and the metrics get worse for no visible reason.

**Not every person is a target**

Classes 7, 8 and 12 (static person, distractor, reflection) are people but are not scored. Training on them teaches the detector to fire on mannequins and reflections.

Both are handled in `mot/data/mot16.py` and covered by unit tests. We wrote the tests first because neither failure produces an obvious symptom.

# Data Augmentation

MOT16 gives us few scenes, so the model can memorise those scenes instead of learning what a pedestrian looks like.

### Detection

- Horizontal flip, boxes mirrored to match
- Colour jitter: brightness, contrast, saturation

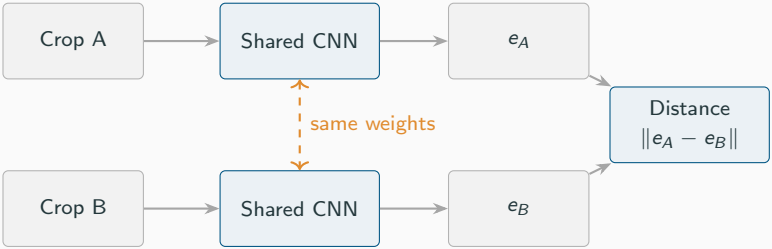
### Re-ID crops

- Random crop and horizontal flip
- Random erasing, simulating occlusion

**What we leave out**

No vertical flips and no large rotations. Pedestrians are upright in every frame of every sequence, so an upside-down person is a pose the model will never see at test time.

# Re-identification: a similarity model, not a classifier



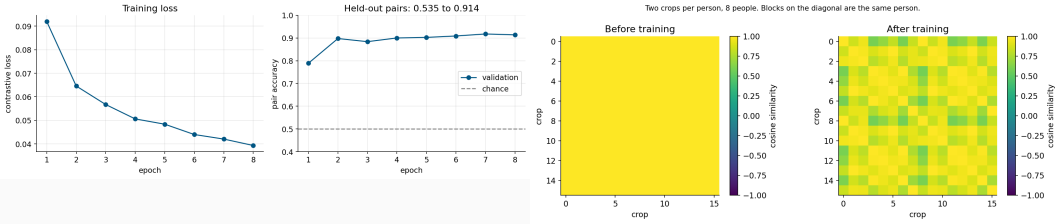

Four convolution blocks, adaptive pooling, then a 128-dimensional L2-normalised embedding. 1.5 M parameters, so it trains in minutes.

**Contrastive loss**

$$L = \tfrac{1}{2}\left[y\, d^2 + (1-y)\max(0, m-d)^2\right]$$

Matching pairs are pulled together, non-matching pairs pushed past the margin $m$.

# Training the similarity model




Eight epochs on crops cut from MOT16 ground truth. Pair accuracy rises from 0.535, which is chance, to 0.932 on held-out pairs. The matrices show the same crops before and after: the blocks on the diagonal are the network learning who is who.

# A problem we had to resolve

**The course slides give four different crop sizes**

Training is specified at $16 \times 16$, inference at $24 \times 24$, retrieved crops at $30 \times 16$, and the reference `Siamese_Network` code uses `nn.Linear(128*22*22, ...)`, needing roughly $102 \times 102$. No single size satisfies all four.

## What we chose

- $128 \times 64$, the resolution Market-1501 normalises its own images to
- `AdaptiveAvgPool2d` instead of a hardcoded flatten

## Why

- $16 \times 16$ cannot resolve clothing, which is the signal re-identification relies on
- Adaptive pooling accepts any input size, so the choice is not baked into the architecture

# How we measure success

## CLEAR-MOT metrics

$$\mathrm{MOTA} = 1 - \frac{\mathrm{FN} + \mathrm{FP} + \mathrm{IDSW}}{\text{ground truth boxes}}$$

- **FN:** a person we failed to detect
- **FP:** a box with nobody in it
- **IDSW:** the ID changed on one person

**MOTP** is the mean IoU of matched boxes, so it measures how tightly the boxes fit.

## Why ID switches matter

A missed frame can be recovered on the next one. An ID switch means we have confused two people, which is the failure the re-identification component exists to prevent and the one the rubric grades.

# Results: first end-to-end run

MOT16-02, 600 frames, on a Colab T4. COCO-pretrained detector, **no fine-tuning yet**; re-identification trained for 8 epochs on MOT16 crops.

## Against the rubric

One real pedestrian is held by a single id for **218 of the 230 frames they appear in (94.8%)**, before any fine-tuning.

But the **median across 44 people is 22.4%**. One person is tracked well; most are not.

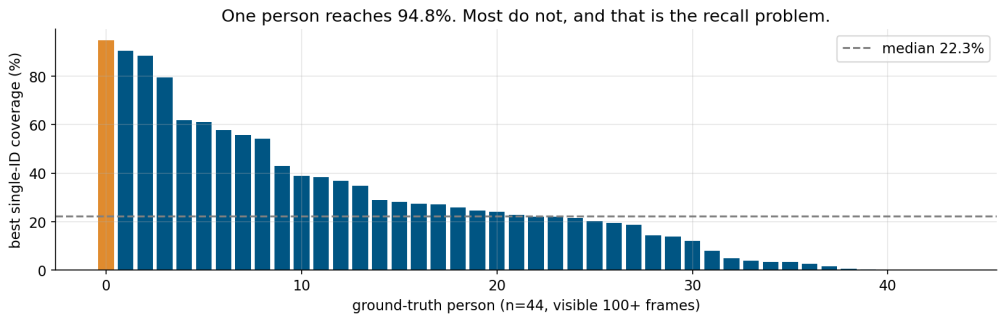
Measured against the ground-truth person, not whichever track lived longest. Re-ID accuracy 0.535 → 0.932, 4.0 FPS.

## Whole sequence

| | |
|---|---|
| MOTA | 0.232 |
| MOTP | 0.792 |
| True positives | 7,004 |
| False positives | 2,713 |
| False negatives | 10,829 |
| ID switches | 151 |

MOTP 0.792 means the boxes are tight. MOTA is held down almost entirely by false negatives.

# Coverage across every person




Best single-ID coverage for each of the 44 people visible for 100 frames or more. One person is tracked almost perfectly; the median is 22.4%. Closing that gap is a detection problem, not a tracking one.

# Does re-identification help? Not yet, and we know why

## Ablation, 600 frames

| | with | without |
|---|---|---|
| MOTA | 0.232 | 0.232 |
| MOTP | 0.792 | 0.792 |
| IDSW | 151 | 148 |

Appearance changes nothing.

## Detection threshold sweep, 150 frames

| thresh | MOTA | recall | FN | IDSW |
|---|---|---|---|---|
| 0.50 | −0.006 | 0.509 | 1,676 | 80 |
| 0.60 | 0.105 | 0.490 | 1,741 | 72 |
| 0.70 | 0.190 | 0.455 | 1,860 | 34 |
| 0.80 | **0.268** | 0.400 | 2,049 | 15 |

## What this tells us

False negatives outnumber ID switches seventy to one, and lowering the threshold lifts recall only 0.40 to 0.51 while false positives quadruple. Recall is capped by people the COCO model never finds: small, distant, occluded. **Detection is the bottleneck, so association has nothing left to disambiguate.** That is the case for fine-tuning, and it is what we do next.

# Demo




`demo_target_clip.mp4`, 8 seconds. Target in colour, every other track in grey.

# Timeline to the final

| Phase | Work | Owner |
|---|---|---|
| Done | Pipeline, parsing, re-ID, metrics, CI | All |
| Done | Baseline run, ablation, demo video | Sanjana |
| Next | Detector fine-tuning (recall is capped) | Ben |
| Next | Re-ID training on real data | Tina |
| Next | Augmentation tuning | Geethika |
| Then | Threshold sweep and ablations | Sanjana, Ben |
| Final | Report, final video, presentation | All |

Code: `github.com/ben-blake/cs5567-project`